

Second Edition

RESEARCH METHODOLOGY IN APPLIED ECONOMICS

Organizing, Planning, and Conducting
Economic Research

Don Ethridge



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The Research Problem and Objectives

A problem well-stated is a problem half solved.

John Dewey

A problem is a perceived difficulty—a difference between what is (or perceived to be or exist) and what might be, which is inherently normativistic. A research problem¹ is a discrepancy between what is known or understood and what is needed or desired to be known or understood. The discrepancy (problem) may concern disciplinary knowledge, subject-matter knowledge, and/or problem-solving knowledge. Research problems are different from, for example, decision or action problems. Research problems address the need to know, perhaps prescribe, whereas decision problems address the need to decide something or take an action. For example, public policy makers face the need to decide among alternative public policies about many things. In other words, they face policy problems, and business people and other managers face business and management problems that require decisions. We all face decision problems, and we all face action problems. We have problems of choosing among alternative courses of action or things to do. Research may provide knowledge and offer prescriptions, but it cannot make a decision and implement it.

We may view problems and problem solving as being of three interrelated types. Once we know what the overall problem is, it may be segregated into three types of problems, or three steps in problem solving. First, there is a problem of obtaining information that is relevant to the overall problem and determining what that information reveals regarding the overall problem and its potential solutions. In its most basic form, this is the research problem component of the overall problem. Second, there is the problem of examining the alternative approaches to the overall problem and choosing among these approaches. This is the decision problem component, and it often relies on results of studying the research problem component. The third problem component is the action problem. Once the decision is made, then how to implement the decision (take action) must be determined.

Our concern in research methodology lies specifically with research problems. We do not take the position that the other types of problems are not important. Our need to recognize the other types of problems arises from the fact that the research problems, issues, and activities do not exist completely separate from the decision and action problems. Conversely, the need exists to maintain a clear distinction between research problems and the other types to avoid the research activity addressing nonresearch matters. This distinction arises at several points in this chapter.

Various classifications of problems exist. For example, Northrop (1959) identifies four kinds of intellectual problems, those of (1) logical consistency; (2) empirical truth; (3) fact; and (4) value.² Back (no date) classifies *research* problems as problems of either intellectual or practical difficulties, suggesting that both disciplinary and applied research problems are important. It also may be noted that meaningful research problems (i.e., those that address an answerable question) and relevant research problems (i.e., those for which the answers are crucial to resolving intellectual or practical difficulties) are related but not the same. For example, some problems may be of intellectual curiosity and capable of being addressed in a research context (meaningful), but have no immediate relevancy to solution of a decision or action problem. This situation occurs more often in disciplinary research than in applied research.

The remainder of this chapter is organized as follows. A distinction is made between research and decision and action problems, followed by a brief consideration of the importance of the problem specification. The researchable problem is then differentiated from the problematic situation, then the problem statement as a part of the research proposal is considered. The use or role of data in the problem specification and a discussion of objectives are presented. The chapter is concluded with some operational suggestions for developing written problem statements and research objectives.

RESEARCH VS. DECISION PROBLEMS

Decision and action problems (and choices about how to respond to them) often go together. We first make a decision, then act on it. Research problems and decision problems also may be closely related, especially when subject-matter and problem-solving research are involved. Disciplinary research is more likely to be directed at problems that are not directly oriented to a specific application or need for individuals or groups to make a decision or take an action. Disciplinary problems are of interest primarily within the discipline and are more likely to be oriented to the generation of knowledge without a specific decision/action purpose (although this generalization has notable exceptions, e.g., much of J. M. Keynes's disciplinary theoretical work apparently had specific policy problems at the forefront). Applied research problems more readily recognize the potential extensions and applications of research that addresses the identified problem. When research

leads to a decision or action, it is potentially important to keep the research activity and the view of the research problem separated from the decision and implementation activities. The reason for this is to maintain as much neutrality as possible in the research process. Keeping them separated helps researchers maintain an objective state of mind.

To help clarify this differentiation, consider the model of problem solving presented by Glenn Johnson (1976, p. 226; see figure 6.1 below). This model illustrates the process that would include pragmatic problem-solving research. The distinction is less a problem in, for example, disciplinary research because disciplinary research is less likely to address the types of problems/issues that require decisions. Here we see the process being initiated with a problem definition, progressing to analysis, which is likely to include a conditional prescription, then someone's decision and implementation of that decision. Also represented is the use of normative and positivistic knowledge and the pragmatic interdependence between them as they impact on each step of the process. The dashed line inserted between analysis and decision in Johnson's model is this author's insertion to separate the research (information-generating) activities from the rest of the problem-solving process. Note that the feedback between the decision and action activities and the analytical activities keeps the analytical closely connected to the decision problem. This separation of research from decision and action presented here is not considered useful, or even valid, by some economists, however. Edwards (1990), for example, sees the decision process as part of the research process. He also recognizes the danger that "when a research economist becomes identified with a certain prescription, a following (or opposition) may be attracted for ideological reasons rather than for the compelling conclusions of the research process" (p. 33). Johnson (1976) may not agree with this distinction within his model. The perspective advocated here is that to mix the research and decision aspects may fragment the researcher's attention, and it may subconsciously influence the outcome of the research.

The admonishment to keep research problems separate from decision problems is not an argument that decision problems are somehow not relevant or are inferior. In fact, the decision problem, and the best answer we can get for it, is often what research users of problem-solving research are most interested in. For that reason among others, we want to keep the knowledge or information on which the decision is based as personally objective as possible. One way to foster that objectivity is to keep research activity focused on problems of knowledge, information, and understanding.

THE RESEARCH PROBLEM IDENTIFICATION

The most critical part of all research projects or plans is the identification of the problem addressed by the research. The problem is the focus of both applied and basic research activity. The reason (justification) for the research inquiry is the

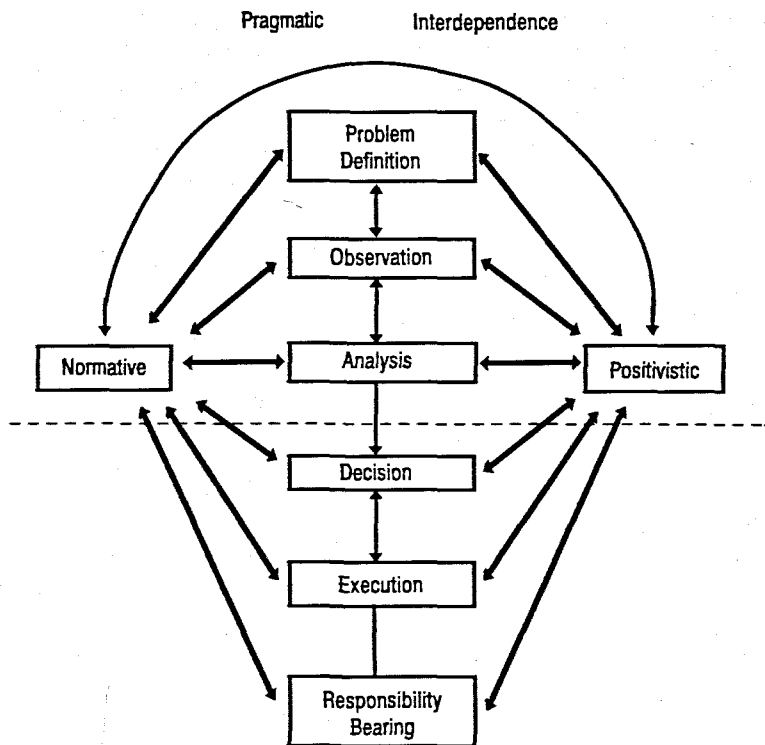


Figure 6.1. Steps in the problem-solving process (Johnson, 1976, p. 226).

research problem. A research problem may be disciplinary, subject-matter, or problem-solving oriented, or it may be any combination of the three. In an operational mode, the problem identification and explanation affect the quality, usefulness, effectiveness, and efficiency of the research activity more than any other part of the research plan. Also, this part of the research plan is often the most difficult to formulate. This observation is based on the author's research and writing experience, working with numerous graduate and undergraduate students on their research projects, and reviewing submitted journal articles, monographs, research bulletins, and the like. Developing a clear understanding of a particular research problem and explaining it so that others readily grasp the problem is difficult. Yet it is the *reason for the research being undertaken*, and the written problem description is the only credible evidence that a clear understanding of the issue has been achieved.

One additional point before moving to more substantive issues: the individual research project addresses the problem, but the problem exists independent of the research. Researchers, and students trying to engage in research, can become

confused by failing to make this distinction. When one wants to know something, that knowledge addresses some definable need if it is to be heeded by anyone, particularly when the research is not strictly disciplinary. Some of us may value knowledge for its own sake, but subject-matter and problem-solving interests have more practical needs. But even in organized disciplinary research, if the problem will not stand on its own, the proposed research is not likely to be taken seriously.

RESEARCHABLE PROBLEM VS. PROBLEMATIC SITUATION

Within the context of a specific research plan, we are most interested in the *specific* research problem that is the point of the research—that is, the researchable problem. The *researchable problem*, put another way, is the very specific problem at which the objectives of the research are directed. It is that particular lack of knowledge or information that constitutes the felt difficulty we attempt to identify, justify, and explain.

An effective way to understand the researchable problem may be to contrast it with the *general* problem, or what Williams (1984) and Andrew and Hildebrand (1982, pp. 19, 20) refer to as the problematic situation.³ The *problematic situation* is a broad problem area or set of problems. Problematic situations are large enough in their scope that the issues embodied in them have many dimensions. Obtaining reliable knowledge and information on all dimensions often requires many years of concentrated analysis. For example, the world food (or hunger) problem is a problematic situation. Its dimensions include low incomes in developing countries, high birth rates in selected countries, national policies that discourage food production, low levels of education in some countries, restrictive trade policies, and political instability that discourages capital investment in some countries, just to name a few. Other examples of problematic situations are: environmental pollution, the balance of trade problem, poverty, affordability of health care, unemployment, and government inefficiency.

Problematic situations are not researchable within the confines of individual research efforts or projects. But specific components of those problematic situations are researchable when they are broken into distinct components. These components are confined in scope and complexity and support a set of research objectives that can be achieved within given resource constraints. This does not say that we must be satisfied with partial knowledge, but individual research efforts must be achievable in order to accomplish anything. The way we accumulate more general understanding is to put knowledge gained from specific research projects together, each having made its own unique contribution to the bigger picture, so that a more general understanding emerges. We may think of a given problematic situation as constituting a picture puzzle of interlocking pieces. A specific research project may provide information that forms one or two of its individual

pieces. Through research programs—a series of planned, coordinated projects—we begin to fill in enough pieces to see the general picture, although the broader understanding may require a degree of extrapolation/interpolation and/or induction. But this is fundamentally the way that scientific knowledge grows. Perfect definitive knowledge, produced quickly, is a noble goal, but it is elusive.

THE PROBLEM STATEMENT(S)

Because the researchable problem is always linked to a more general set of problems, it is recommended that the problem statement in a research proposal be separated into two parts: the general problem (or problematic situation) and the specific problem (or the researchable problem). Note that, as was pointed out in chapter 5, the researchable problem also must have some means of being addressed. There must be data, analytical procedures, and other elements with which to address the problem. Thus, merely being specific is not sufficient. Note also that the criterion for the specific problem is not that it be “small,” but rather that it be precisely specified and capable of being addressed with available resources.

The general problem statement in a research proposal is to provide the background or the setting for the researchable problem. It gives the larger picture into which the researchable problem fits—the context of the problem that the proposed research addresses. A useful rule of thumb is to introduce the problem in generalities, then progressively narrow the focus (your own attention and the attention of readers) to more specific, precise problems on which information is needed. One may choose to initiate the general problem description with a very broad problematic situation, particularly when the readers of the proposal may not already be knowledgeable on, say, an economic perspective of the broad picture. Alternatively, one may initiate the general problem description with a more confined treatment of the problem setting, particularly when the readers are likely to be already familiar with the disciplinary aspects of the problem. The problem definition, as with any written communication, needs to be tailored for the expected audience.

To illustrate, consider a typical situation encountered when a student from another country is conducting research in the United States, but the focus of the analysis is on the student's home country. In this situation the student may be defining the problem setting for his or her graduate committee, made up largely or partially of faculty members who are not familiar with the issues of the country being studied. In this type of situation, it may be advisable to provide some description of the economic, social, political, and/or institutional conditions in the country of study. This understanding may be necessary for the committee members to grasp the relevant dimensions of the problem setting. On the other hand, if the topic is on a problem, say, within the United States, and the committee is

already familiar with the economic, political, and institutional frameworks, those matters may need less extensive discussion and explanation.

Consider, for example, the problem definitions in the proposals in appendices A, B, and C. The general problem background in the government agency proposal relating to effects of a type of policy structure on the part of some governments on various countries (appendix A) emphasizes governments' incentives and reasons for intervening in markets, both domestic and international, and some of the general effects of those interventions. It establishes the background against which a particular set of policy impacts will be analyzed. The dissertation proposal on pharmaceutical mergers (appendix C) explains mergers in terms of types, reasons, impacts, and trends; it addresses mergers in general as a prelude to laying the groundwork for identifying the (researchable) issues of mergers in the pharmaceutical industry. The thesis proposal on the impacts of HIV on the Russian economy (appendix B) describes the general social and economic situation in Russia—a sort of “baseline”—to frame the economic problems associated with the expanding incidence of HIV in the complex mix of forces affecting the economy.

The general problem statement sets the stage for the specific, researchable problem specification. The specific problem continues to narrow the perspective of the problem to a set of issues on which knowledge or information is needed. In this process, it is necessary, but not sufficient, to make clear what knowledge is needed. The reasons the knowledge is needed, by whom it is needed, and the potential purposes it could serve should also be clear. You must try to guide your readers' thinking in a way that they perceive the problem and need for information the same as you perceive them. In fact, if the purpose of the problem statement is to justify the objectives of the research, then the test of effectiveness of the problem definition is that readers unhesitatingly accept the research objectives. If the readers' response to the research objectives is “Of course, that's what is needed,” the problem statement has effectively achieved its purpose.

The five guidelines listed below are presented as a sort of checklist to aid thinking about effective problem statements that lead to researchable problem specification, whether the problem/issue/question is disciplinary, subject-matter, or problem-solving.

A. *The researchable problem must be sufficiently specific (confined) that it can be addressed with available resources.* These resources include time, researcher expertise, data, tools (e.g., computer capacity), support personnel, and financial support. The magnitude of the researchable problem is always dependent to some degree on the resources available. What constitutes a feasible researchable problem for a researcher with a large grant, a staff of research personnel, previously accumulated databases, and abundant computer capability is obviously different from a feasible researchable problem for a graduate student without those resources. A disciplinary researchable problem must have the theoretical or methodological issue precisely identified, and the researcher must have sufficient capability to attempt the theoretical and/or methodological work. One

should remember, however, that the quality of a problem specification is not dependent on how large the researchable problem is, but on how *clear* it is. As a rule, the large (broad) researchable problems are usually less clear than the confined, well-focused ones.

B. *The dimensions of the problem should be described in objective (neutral) rather than subjective (advocative) terms.* Avoid "value-judgment" language. Although the problem definition is inherently normative, discipline yourself to be precise in your description and subjugate your private values and biases. The language you choose to use may affect both the perception of how objective/neutral your research is and the reality of your neutrality. For example, if you state, "Wheat producers in the European Economic Community are overpaid," the statement implies personal subjectivity, whereas "Wheat producers in the European Economic Community are subsidized" avoids that subjective connotation. It may also help you to be more precise in your own thinking about the issue. The existence of subsidies can be substantiated (e.g., "Internal wheat prices are 2.5 times the world market price for wheat"). Whether they are overpaid is a personal judgment; how much of a subsidy constitutes "overpayment" is a matter of opinion.

Avoiding subjective language in defining the problem may prevent the discussion of the problem focus from drifting into areas that have no answers. Also, avoiding unnecessary or unneeded descriptors, particularly those that express an opinion, is also advisable. For example, "Unfortunately, 20% of small businesses go bankrupt each year" is a statement for which the addition of *unfortunately* adds no substantive understanding; it connotes an emotional appeal.

C. *The problem must be described sufficiently that other people (the audience) comprehend it.* This applies to both the general and the specific problem statements.

D. *The researcher's perception of the problem may be derived from an intuitive level, but it should be developed to a logical level.* It is only in this manner that the researcher and others may really understand the problem. If the description of the problem is not developed logically to a point that others grasp it, then the problem is not well defined. Emotional statements fail to offer evidence.

E. *The inquiry may be initiated with a research objective—something that you want to determine—but the problem explanation should be developed to provide the justification for the objective.* If you cannot explain the problem that is the reason (justification) for the objective(s) of the research, then either you do not clearly understand the problem yourself, or there is no problem.

USE OF DATA

In the processes of understanding and describing problems, several important components may be involved. We have noted that our perceptions, conditioned by personal experiences, are likely to be involved. Theory also plays an important

role in understanding the problem, as will be discussed in chapter 8. Yet another component of understanding the problem, be it a problematic situation or a researchable problem, is data (Edwards, 1978, 1990).

Data, and the process of collecting them, can affect how one perceives a problem and one's awareness of a problem. Data often are instrumental in revealing problems that might have otherwise gone undetected. Consequently, it is usually advisable to seek out and examine relevant data in the process of developing the problem statements. If your research topic includes employment considerations, go to the library and locate appropriate data series on employment statistics. If your topic includes trade in automobiles, locate data series on automobile production, exports, and imports, perhaps in different countries. You may be surprised at what the data reveal, and how the selected inclusion of some of those data in your problem definition can foster the readers' comprehension of the problem. Edwards (1978, p. 31) may best state the case:

In the descriptive process, people arrange the available data to tell a story and to illustrate a problem. The person who understands the ultimate social problem must sift through, classify, aggregate, and rearrange the data and reduce large volumes of miscellaneous facts to manageable information. This interaction between unorganized data and social problems results in meaningful descriptions. It carries us beyond vaguely felt needs toward precise conceptualizations of a problem and provides for measures of its nature and extent.

Descriptive research, or the process of synthesizing information, often includes the interpretative use of data that may be included in problem definitions, as discussed in chapter 2. Put another way, defining research problems often includes conducting descriptive research, that is, synthesis. Note, for example, how the proposal in appendix C uses data to show changes in market structure in the pharmaceutical industry.

OBJECTIVES

The objectives specified in the research proposal address the specific, researchable problem. They specify what the research project proposes to accomplish (do, achieve, determine, estimate, measure, evaluate, etc.). Objectives are usually best specified in the form of (1) a general objective; and (2) a set of specific objectives (refer to the objective statements in appendices A–C). Although the statement of objectives is often the shortest part of any research proposal, it remains the centerpiece. The other parts of the proposal support and give credence to the objectives.

The *general objective* is the central objective of the research study. It states the main purpose of the research. It also should flow directly from the statement of the researchable problem. For example, if the specific problem is that little is known about the response of toxic waste emissions into the air from government

finer for toxic waste emissions specified by particular legislation, then an appropriate general objective might be to determine how the legislated schedule of fines has affected, or would affect, the concentration of toxic waste emissions in the northeastern United States. The more succinctly and precisely the general objective can be stated, the better its function is served. A useful guideline is that general objectives should be stated in one sentence.

The *specific objectives* constitute a set, perhaps a list, of subobjectives that contribute to the general objective. The achievement of each specific objective contributes something toward achieving the general objective, and the primary objective is achieved in total or in part by achieving the set of specific objectives. Sometimes additional analysis that involves combining results from specific objectives is needed as well.

Continuing with the above example of a general objective, appropriate specific objectives might be to: (1) identify the legislated schedule of fines specified for toxic waste polluters; (2) estimate the responsiveness of toxic waste producers in the northeastern United States to various levels of fines (and possibly to the standards of enforcing the fines); and (3) determine the expected response of toxic waste emitted to levels of fines other than those specified legislatively. Note that the researcher must have already done enough background work to expect that the objectives are achievable (the problem is actually researchable). This may include conceptual and preliminary modeling work, exploration of data sources, and knowledge of prior studies related to the topic.

DEVELOPING PROBLEM AND OBJECTIVE STATEMENTS

As is usually the case in creative matters of this type, there are no specific, unassailable rules on how to develop the statement of the problem and the objectives in a research proposal. Furthermore, what works for you in one set of circumstances may not work the same again when the situation varies. The suggestions that follow are derived from having struggled with the process and having worked with students who were struggling to develop clear, effective, efficient statements of problems and objectives. You may start with either a broad subject-matter area in which you are interested in conducting research or a specific researchable problem or a general objective, as noted above. Suggestions are segregated on the basis of the way in which you initiate the research inquiry.

If you begin with a broad area in which you want to conduct your research, remember that the task is to get that research interest narrowed to a researchable problem. Start by segmenting the problem area into its logical components, then continue to segment those into smaller parts (issues, questions, etc.). Unless you have a natural affinity for the process or are very lucky, you will likely need to work through multiple iterations of the process. Discuss the questions, needs, issues with peers and advisors; explain what you already understand and try to explain where you are stuck. Sometimes the act of verbalizing your thoughts

gives more clarity to them. Explore all aspects of the problem, and at all levels of specificity. Remember, however, that you probably will not want to include everything you have thought of in your written statements. Your written statements need to follow a direct, linear train of thought. Getting sidetracked will only confuse your readers and evaluators. Follow a progression in your explanation—sequentially narrow the reader's focus to a progressively specific problem.

If you start with the opposite perspective—you already have an idea of your researchable problem (or general objective)—then you must develop and explain the more general dimensions of it so that readers and evaluators can also understand the context and setting of the researchable problem. The task is to “back-track” through the logical sequence leading to the justification of the researchable problem. Starting from the statement of the researchable problem, ask yourself why the researchable problem exists (why the knowledge is needed), to whom it is a problem (who needs the knowledge or information), and how it became a problem. Proceed by asking these questions at the next level of generalization, and proceed to the level of generalization needed for a starting point. Remember that you are searching for a point at which to initiate your description of the general problem or problem situation. Your focus can become too general—too elementary for the audience. Once you can identify a starting point for the general problem statement, then reverse the process and fill in the logical sequence in the same manner as above.

Once the problem has been defined, framed, and justified, and the general objective therefore identified, the specific objectives become a matter of identifying an appropriate set of subobjectives that leads to the general objective. Ask yourself what component parts must be determined or understood in order to achieve the knowledge required for the general objective. For example, if determining an economic optimum for a procedure, process, or policy is a general objective, then determining the costs and benefits associated with the procedure, process, or policy are likely to be specific objectives. You may find that the identification of specific objectives is rarely mentally or conceptually demanding, but it requires attention to details.

A note of caution on specifying specific objectives: avoid confusing objectives with methods. The specific objectives address *what* is needed to achieve the general objective, not *how* the general objective (or specific objectives) will be achieved. Students often want to state a specific objective as “To construct a model of . . .” when constructing a model is a means of achieving an objective (a method or procedure), not the objective itself, especially in subject-matter and problem-solving research. Granted, the two may go together, but what perspective has the researcher lost by confounding means with ends? Developing new methods *may* constitute an objective, but that situation identifies a purely disciplinary activity. Thus, the separation between objectives and procedures may, on occasion, be violated, but the instances when a model is a technically correct objective are limited. Keeping objectives separated from the methods and procedures remains the best *general* approach.

Another note on specifying and writing problems and objectives: do not expect this exercise to be a straightforward, once-through endeavor. Unless you are experienced and already have done extensive research in the topic area of the proposed research, expect to rotate back to the problem statement and objectives repeatedly as you gain more background and perspective: for example, as you proceed through more literature search, conceptual thinking, and data. Consider that the problem statement presents a perspective, interpretation, or point of view about a problem. It is therefore not unusual that you would alter your perspective, and your explanation of it, as you gain more understanding of matters that are necessarily related to the problem. Do not be surprised if your understanding and description of the problem are altered substantially from the time you initiate your research project until you complete it.

Most researchers find that developing a logical, clearly written statement of the problem and objectives is the most difficult and demanding aspect of proposal writing, if not of the entire research process. It requires clearer thinking and more communication skills than perhaps any other part of the research proposal. Yet proposals may succeed or fail most often based on their problem specification than on any other single component. Few reviewers feel comfortable in judging the substantive procedural aspects of proposed research unless or until they understand the problem. The technical aspects of your proposed research are unlikely to receive serious consideration without a clear problem description. A good problem statement will not, however, rectify incomplete or poorly thought-out procedures.

It is difficult to overemphasize the importance of peer and advisor review and input in the process of developing problem statements and statements of objectives. Interacting with peers and advisors, both orally and through written drafts, is not only a device for increasing the efficiency of our communications, it is necessary for many of us to clarify our thinking. You may find it useful at this point to review the suggestions on writing in chapter 5.

SUMMARY

This chapter considers the definition of research problems and their associated objectives from the perspectives of both formulating them and writing them for a research proposal. A distinction is made between research and decision problems, and the reasons for differentiating between them in the research process are discussed. Research problems are often defined by the need to solve decision problems. Conversely, the solution of problem-solving and subject-matter research problems may, and hopefully often do, provide the knowledge or information with which decision problems are addressed.

The research problem is the focal point of research activity. We can visualize it as having two components: the *general* problem, or problematic situation, and the *specific* problem, or the researchable problem. It is the researchable problem delineation that is most important for focusing the individual research project or

activity. However, specification of the more general problematic situation framework is necessary to place the researchable problem in context.

Objectives are also presented as needing to be specified in general and specific terms. General objectives identify the overall intent of the research project, while specific objectives identify the subparts that will contribute to achievement of the general objective. Problems with confusing objectives with methods and procedures are discussed.

Guidelines for writing the problem statements are provided. These include considerations of specificity (confinement of the researchable problem), language use, and use of empirical data to demonstrate problems. Guidelines on writing objectives are also provided.

SUGGESTED EXERCISES

1. Obtain a research proposal (e.g., from a student, faculty member, or research organization). Read the problem and objective statements. Can you identify the problematic situation, the researchable problem, the general objective, and the specific objectives? Explain.
2. Select two journal articles from recent issues of economics journals, preferably different journals. Explain the extent to which each addresses research objectives and decision objectives as they are explained in this chapter. Can you identify the problematic situation and the researchable problem in each? Can you identify subjective language in the problem statements (or in any other part of the articles)?

NOTES

1. A *problem*, as viewed in this context, may also be labeled *issue* or *question*. Readers may substitute these labels as desired.
2. Back (no date) noted that it is curious that Northrop omitted problems of method dealing with how to generate reliable knowledge.
3. Breimyer (1991, p. 247) attributes the original delineation of the problematic situation as separate from the research problem to Salter (1942).